Here is the final, comprehensive **Technical Specifications & UI Brief** to hand over to your development team. It combines the system requirements with the strict focus on a high-end, high-performance visual design.

📋 Comprehensive Developer Specification Document

Project Title: Official Web Portal & Leo Academy Engine for Leo District 306 D3
Core Directive: This is a frontend-centric web application where **visual design, premium aesthetics, and smooth animations** take absolute priority. However, all visual polish must achieve clean performance metrics, rendering near-instantly on both desktop and mobile devices.

🎨 1. Frontend UI/UX Design System (The Highest Priority)

The interface must feel luxurious, modern, and deeply engaging. It should avoid a static, text-heavy appearance by utilizing fluid movement and micro-interactions.

Color Architecture

- **Dominant Backgrounds (80%):** Deep, premium dark tones. Use a mix of true black (#000000) and rich charcoals (#0B0C10, #121212) to create depth.
- **Accents & Details (20%):** Metallic Golds (#D4AF37, #C5A059). Use gold purposefully for borders, active components, action buttons, and focal elements. **Never oversaturate.**

Animation Guidelines (Smooth Motion, High Speed)

To keep the site fast on mobile, **do not use heavy, bloated JavaScript animation libraries.** Instead, leverage hardware-accelerated CSS properties:

- **Layout Reveals:** Implement sleek fade-and-rise entries for sections as the user scrolls. Use the native browser **Intersection Observer API** paired with CSS transitions (opacity, transform: translate3d) for zero-lag rendering.
- **Micro-Interactions:** Custom buttons should feature fluid transitions on hover (e.g., subtle gold glow expansions, elegant gradient slides, or smooth color-inversions).
- **Zero Layout Shifts:** All animated elements must have pre-defined spacing boundaries to avoid jarring shifts while the page loads.

Mobile-First Performance Metrics

- **Layout Framework:** Bootstrap 5, optimized using clean utility classes rather than deep custom element overrides.
- **Asset Performance:** All graphics and photography must use highly compressed, next-gen **WebP** formatting. Set font configurations to font-display: swap to ensure text renders immediately without waiting for server responses.

🖥️ 2. Public Portal Page Layouts

1. Home Page

- **Hero Unit:** A full-bleed, cinematic dark background section presenting the current District President's theme and high-impact project photography.
- **Impact Tracker:** Active count indicators mapping total clubs, current membership numbers, and served volunteer hours.
- **Highlights Stream:** A clean card layout detailing recent press releases and key district milestones.

2. About Us

- **The Pillars:** Engaging visual presentation of the Leadership, Experience, and Opportunity framework.
- **History:** A clean narrative layout mapping out the history of District 306 D3.

3. Leadership Directory

- **Structure:** Clean, asymmetrical card grids mapping out the District Executive Committee, Directors, and Region/Zone Chairpersons.
- **Details:** Unified profile photographs, official titles, and active communication links.

4. Clubs of the District

- **Interface:** A searchable archive allowing users to sort local clubs by Region, Zone, or classification (School vs. Community-based).
- **Profiles:** Minimalist summary cards presenting the current Club President, scheduled meeting timings, and direct social handles.

5. District Calendar

- **Interface:** A monthly calendar UI displaying upcoming timelines.
- **Categorization:** Color-coded tag identifiers separating mandatory Council Meetings, District Projects, and local Club Events. Include an "Add to Personal Calendar" button.

🧪 3. The "Leo Academy" Engine Specification

A lightweight, frictionless certification workflow managed via session states instead of traditional authentication systems.

[Candidate Entry Form]



Page Mechanics (quiz.html)

- **Automated Timer:** Execute a prominent JavaScript visual timer in the browser window. If the counter reaches 00:00, force an automatic form submission bypass to save current progress.
- **Submission Endpoint (POST /academy/submit/):**

1. Read the active validation key out of the Django session.
2. Grade selected answers directly against the correct choice fields.
3. If the final score meets or exceeds the designated pass percentage, flag is_passed = True.
4. Write the results entry to the LeoCandidate model.
5. **Mandatory:** Flush the active Django session immediately to completely restrict secondary attempts under that NIC.
6. Render the final outcome display.

🗄️ 4. Data Layer & Administration (Django Models)

```python
from django.db import models

class Club(models.Model):
    name = models.CharField(max_length=150, unique=True)
    region = models.IntegerField()
    zone = models.IntegerField()
    club_type = models.CharField(max_length=20, choices=[('School', 'School-based'), ('Community', 'Community-based')])
    president_name = models.CharField(max_length=255)
    meeting_details = models.TextField()
    contact_email = models.EmailField()
```

```
    facebook_url = models.URLField(blank=True, null=True)

class Event(models.Model):
    title = models.CharField(max_length=200)
    description = models.TextField(blank=True)
    start_time = models.DateTimeField()
    end_time = models.DateTimeField()
    category = models.CharField(max_length=30, choices=[('Council', 'Council Meeting'),
('District', 'District Project'), ('Club', 'Club Project')])
    is_draft = models.BooleanField(default=True)

class QuizQuestion(models.Model):
    question_text = models.TextField()
    marks = models.IntegerField(default=1)

class QuestionChoice(models.Model):
    question = models.ForeignKey(QuizQuestion, on_delete=models.CASCADE,
related_name='choices')
    choice_text = models.CharField(max_length=255)
    is_correct = models.BooleanField(default=False)

class LeoCandidate(models.Model):
    nic = models.CharField(max_length=12, unique=True)
    full_name = models.CharField(max_length=255)
    club_name = models.CharField(max_length=150)
    score = models.IntegerField()
    is_passed = models.BooleanField()
    submitted_at = models.DateTimeField(auto_now_add=True)
```

Django Admin Setup (admin.py)

- Use admin.TabularInline for QuestionChoice to allow the IT team to add questions and their choices on the same page.
- Implement django-import-export on the LeoCandidate interface to provide a single-click Excel export tool for tracking candidate logs.

🚀 5. Production Infrastructure (Railway Deployment)

1. **Environment Variables:** Manage configurations using production keys (DEBUG=False, SECRET_KEY, and auto-wired DATABASE_URL).
2. **Server Process:** Include a root level Procfile mapping the runtime layer: web: gunicorn project_name.wsgi --log-file -.
3. **Requirements:** Pin essential application runtimes cleanly inside requirements.txt: django, mysqlclient, gunicorn, and django-import-export.